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School of Computer Science and Artificial Intelligence

ECO TRACK

“EcoTrack: A Smart Plastic Waste Recycling Tracking System”

A centralized digital platform to monitor and track plastic waste from collection to recycling, improving transparency, efficiency, and sustainable waste management.

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Limitations

- **Manual Data Dependency**
  - System relies on manual data entry by users
  - May lead to human errors or incorrect data

- **Lack of Automation**
  - No integration with IoT sensors or smart bins
  - Waste tracking is not fully automatic

- **Limited Real-Time Tracking**
  - Does not support GPS-based transportation tracking
  - Status updates are manual

- **Basic Analytics**
  - Provides only simple charts and reports
  - Lacks advanced data analysis features

- **Scalability Constraints**
  - Designed as a prototype
  - May not handle large-scale city data efficiently

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Future Scope

- **IoT Integration**
  - Use smart sensors to automatically track waste levels
  - Reduce manual data entry

- **GPS Tracking**
  - Enable real-time tracking of waste transportation
  - Improve monitoring accuracy
- **AI & Predictive Analytics**
  - Predict waste generation trends
  - Optimize recycling processes

- **Mobile Application**
  - Develop mobile app for collectors and authorities
  - Improve accessibility and usability

- **User Authentication System**
  - Add login and role-based access
  - Improve security and data management

- **Smart City Integration**
  - Integrate with government systems
  - Scale for large urban environments

- **Advanced Reporting**
  - Generate detailed reports and insights
  - Enable better decision-making